Unit: Unit 8: Quadratics **Name:** ____________
Topic: Vertex Form and Intercept Form **Date:** __________

Advanced (Carolina Reaper)

Q1. Write the quadratic function $f(x) = x^2 + 4x + 4$ in vertex form.
(A) $f(x) = (x + 2)^2$
(B) $f(x) = (x - 2)^2$
(C) $f(x) = x^2 + 4x + 4$
(D) $f(x) = (x + 4)^2$
(E) $f(x) = (x - 4)^2$

Q2. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$. What does h represent?
(A) The y-intercept
(B) The x-intercept
(C) The vertex of the parabola
(D) The axis of symmetry
(E) The maximum or minimum value

Q3. Write the quadratic function $f(x) = x^2 - 6x - 7$ in intercept form.
(A) $f(x) = (x - 7)(x + 1)$
(B) $f(x) = (x + 7)(x - 1)$
(C) $f(x) = (x + 6)(x - 1)$
(D) $f(x) = (x - 6)(x + 1)$
(E) $f(x) = (x - 1)(x + 7)$

Q4. The intercept form of a quadratic function is $f(x) = a(x - p)(x - q)$. What do p and q represent?
(A) The y-intercepts
(B) The x-intercepts
(C) The vertex of the parabola
(D) The axis of symmetry
(E) The maximum or minimum value

Q5. Which of the following is the vertex form of the quadratic function $f(x) = 2x^2 - 4x - 6$?
(A) $f(x) = 2(x - 1)^2 - 8$
(B) $f(x) = 2(x + 1)^2 - 8$
(C) $f(x) = 2(x + 2)^2 - 6$
(D) $f(x) = 2(x - 2)^2 - 6$
(E) $f(x) = 2(x + 2)^2 + 6$

Q6. What is the axis of symmetry of the parabola $f(x) = x^2 - 4x - 3$?
(A) $x = -1$
(B) $x = 1$
(C) $x = 2$
(D) $x = 3$
(E) $x = 4$

Q7. Write the quadratic function $f(x) = x^2 + 2x - 6$ in vertex form.
(A) $f(x) = (x + 1)^2 - 7$
(B) $f(x) = (x - 1)^2 - 7$
(C) $f(x) = (x + 2)^2 - 6$
(D) $f(x) = (x - 2)^2 - 6$
(E) $f(x) = (x + 1)^2 + 7$

Q8. What is the vertex of the parabola $f(x) = 2x^2 - 4x - 3$?
(A) $(1, -5)$
(B) $(-1, -5)$
(C) $(2, -3)$
(D) $(-2, -3)$
(E) $(1, 5)$

Open Ended Questions (FRQ)

Q9. Write the quadratic function $f(x) = 3x^2 - 12x - 2$ in both vertex form and intercept form. Identify the vertex, axis of symmetry, and x-intercepts.

Q10. A quadratic function has its vertex at $(2, -3)$ and passes through the point $(0, 5)$. Write the equation of this quadratic function in vertex form and intercept form.

Chef's Tips

- Allow students to use graphing calculators to check their answers.
- Emphasize the importance of showing all work and explaining reasoning.
- Provide extra support for students who struggle with quadratic functions.